

CertVIC: Separating Visual Responsiveness from Intervention Specificity in Open Vision-Language Models

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Evidence status. This V11 draft reports a real, pilot-scale set of open-model outputs, but it is not a submission-ready empirical paper. The item-validity screen used to form the 91-item intervention set was machine-assisted, the independent second-rater sheet is blank, and the 30-item Spurious V2 follow-up has no model outputs or human labels. Numerical confidence-sequence crossings are therefore reported separately from full scientific claim eligibility.

Abstract

Vision-language model evaluations often reward a response change after an image edit, but responsiveness alone is ambiguous: a model may react to any visible perturbation rather than to the intended semantic intervention. CertVIC separates two paired quantities: responsiveness to edits that should change an answer and specificity under edits that should leave it unchanged. A pilot on three open 7–8B models yields large descriptive intervention gaps on 91 machine-screened ADE20K pairs, with anytime-valid lower bounds above the configured numerical threshold. The specificity control changes the interpretation. On 94 irrelevant-edit pairs, Qwen2.5-VL-7B flips on 12 items (0.1277) and fails the frozen observed-rate gate of 0.10; InternVL2-8B and LLaVA-OneVision-7B flip on 1 and 3 items. All twelve Qwen failures are Qwen-only within the paired three-model comparison, and parser/provenance checks do not explain them. These results motivate a model-dependent specificity hypothesis, not a broad robustness claim. We describe the evidence boundary, the prospective control protocol, and the human and scale validations required before stronger conclusions.

1 Why responsiveness and specificity must be separated

An edited image can change a model answer for at least two reasons. The intended semantic fact may have changed, or the model may simply be sensitive to a low-level perturbation. Conversely, an unchanged answer after a relevant edit may indicate decision inertia even when the model recognizes natural object absence. A useful intervention evaluation must therefore measure both responsiveness and invariance under a matched no-change control.

CertVIC’s primary contribution is an evaluation protocol and estimand, not a collection of execution wrappers. The protocol links a source image, a documented edit, a locked question/answer pair, raw model responses, strict parsing, and a claim gate. Its statistical component supplies a time-uniform uncertainty bound for a predeclared item order. The pilot’s scientific observation is narrower: responsiveness and specificity are empirically separable, and specificity is model dependent on the available control.

Scope relative to prior work. The repository’s related-work matrix identifies counterfactual VQA, visual robustness, consistency testing, and anytime-valid inference as the closest categories, but verified bibliographic anchors are still missing. This draft therefore makes no priority claim. A submission must establish which prior methods measure answer updates without a matched irrelevant-edit gate, and which uncertainty procedures remain valid under continuous monitoring.

2 Paired estimands and statistical decision

For eligible item i , let A_i be original-image correctness and let C_i indicate whether the parse-valid response pair changes for a relevant intervention or remains unchanged for an irrelevant control. The historical C_i does not require the edited answer to equal edited gold; correct semantic update success is reported separately. The descriptive quantities are

$$a = \frac{1}{n} \sum_i A_i, \quad p = \frac{1}{n} \sum_i C_i, \quad \Delta = a - p. \quad (1)$$

Table 1: Real pilot outputs. a , p , and Δ use the 91 intervention pairs. The CS lower bound is a numerical time-uniform bound, not full evidence eligibility. The V1 specificity gate is the frozen observed-rate rule.

Model	n	a	p	Δ	CS LB	V1 flips	Gate
Qwen2.5-VL-7B	91	.923	.176	.747	.364	12/94	fail
InternVL2-8B	91	.923	.099	.824	.441	1/94	pass
LLaVA-OneVision-7B	91	.890	.176	.714	.331	3/94	pass

The historical symbol p denotes consistency, not a p-value. For a no-change control, $F_i = 1 - C_i$ and $\hat{f} = n^{-1} \sum_i F_i$ is the spurious flip rate.

The implementation maps the gap variable to

$$D_i = (A_i - C_i + 1)/2 \in [0, 1], \quad \Delta = 2\mathbb{E}[D_i] - 1, \quad (2)$$

and applies a Hoeffding normal-mixture confidence sequence with a data-independent scale set by the planned horizon. Its time-uniform coverage supports optional stopping for the predeclared order under the bounded conditional-mean assumptions. It does not establish edit validity, use of the intended visual evidence, provider reproducibility, or population generalization. Those require separate gates.

Frozen V1 rule and prospective V2 rule. The V1 decision remains the observed-rate rule $\hat{f} \leq 0.10$; it is not changed after seeing outcomes. For V2, Qwen is the predeclared primary model and the prospective rule uses a one-sided 95% Clopper–Pearson upper bound that must be at most 0.10. A joint three-model statement uses Bonferroni level 0.05/3 per model. Missing or unparseable pairs fail the primary endpoint. Raw and pre-result validity-filtered analyses are both reported.

3 Repository evidence and validity status

The intervention pilot contains 91 item pairs selected from 103 candidates. The source images and masks are from the local ADE20K release; edits include removal, displacement, and occlusion. The predictions are real outputs imported from free Kaggle runs for Qwen2.5-VL-7B, InternVL2-8B, and LLaVA-OneVision-7B. Their exact model commit revisions were not recorded, which limits replication across changing model repositories.

The repository labels this item set “human reviewed,” but its completed review CSV identifies the reviewer as `assistant_visual_review_v1`. The independent second-rater export contains 91 blank rows. V1 therefore treats the model predictions as real observed evidence and the review-derived inclusion decisions as machine-assisted preliminary. This distinction weakens claim eligibility without altering the raw responses or the numerical estimators.

The V1 specificity arm contains 94 paired irrelevant edits per model. All 188 response rows per provider parse as strict yes/no tokens; item IDs, providers, image pairs, and merged zip members match. An objective image audit finds no target-mask overlap, but 20 of 94 difference bounding boxes intersect conservative target boxes. The detectability diagnostic reports symmetric grouped-item AUC 0.7123, below its repository flag threshold of 0.8 but clearly not proof of imperceptibility.

4 Pilot results

The intervention gaps are large for every model in this pilot, and the numerical confidence-sequence lower bounds exceed the configured gap threshold 0.05. Those values are reproducible from the raw paired rows. They do not clear the full policy: the set has $n = 91$ versus the configured minimum 150, lacks completed independent human validation, and Qwen fails the separate specificity gate.

For Qwen and InternVL, every observed answer change on this arm also moves from the original gold answer to the edited gold answer (16/91 and 9/91). LLaVA changes on 16/91 items, but only 13 end on the correct edited answer. This distinction is why p is described as raw response change rather than semantic success.

The natural present/absent control indicates that the models can recognize unedited absence: combined accuracy is 0.897 for Qwen, 0.935 for InternVL, and 0.932 for LLaVA on 369 validation images. Thus the observed intervention behavior is not explained solely by an unconditional tendency to answer that the queried object is present. The control does not by itself validate the edited images.

Paired model comparison. On the same 94 specificity items, the Qwen-minus-InternVL flip-risk difference is 0.117 and the Qwen-minus-LLaVA difference is 0.096. Exact McNemar p-values are 0.0034 and 0.0352; after a three-comparison Holm family the second contrast is not below 0.05. These analyses are retrospective and exploratory. InternVL versus LLaVA has risk difference -0.021 and exact p-value 0.625.

5 Forensics of the twelve Qwen flips

Each Qwen failure changes from parsed “yes” on the original image to “no” on the edited image. The corresponding InternVL and LLaVA pairs do not flip. No missing row, duplicate item/variant key, parse error, provider mismatch, or image-path mismatch was found. Two Qwen failures have conservative difference bounding boxes intersecting the target box, while target-mask overlap is zero for every item. Machine-generated visual categories in older artifacts are diagnostics and are not used as human-validity labels.

This pattern is consistent with model-dependent sensitivity to irrelevant edits, but it does not identify a mechanism. The model revisions, quantization details, and exact processor commits are incomplete in the historical run manifests. A repeat under pinned revisions and a blinded validity review is required before assigning the difference to architecture rather than run configuration.

6 Prospective Spurious V2 and scale decision

The repository contains a stricter 30-item V2 package built from the 94-item source pool. Its objective construction enforces zero target-mask overlap, no target-box intersection, minimum box distance 75 pixels, and a stored detectability-score cap of 0.12. The set was constructed after V1 was observed. Four of the twelve known Qwen failures remain in the set and eight are removed by post-V1 geometry/saliency rules. It is therefore a retrospective stricter-control sensitivity set, not an independent confirmatory sample.

No V2 provider output or real human label is present. Moreover, $n = 30$ is too small for a strong simultaneous three-model upper-bound statement: even zero failures cannot put a one-sided Bonferroni upper bound below 0.10. V2 is useful as a re-encoding/strict-filter sensitivity check, but it cannot clear a confirmatory specificity gate. A larger independently sourced control set containing no V1 items is needed for a broad specificity claim.

Human review must precede result unblinding. Two independent raters receive randomized anonymous pairs without provider outputs or V1 failure labels. They judge target preservation, answer invariance, perturbation acceptability, ambiguity, answerability, and retention. The raw set remains reported regardless of exclusions.

Main-500 remains blocked. It becomes scientifically useful only after the specificity branch is resolved, item selection and replacement are outcome blind, model revisions are pinned, and real human review and objective edit-quality gates pass. A Qwen failure does not require abandoning the project; it changes the paper’s question from uniform specificity to when responsiveness and specificity diverge across models.

7 Limitations and research path

The current evidence has five dominant limitations. First, human item validity is not complete. Second, the sample is small and from one dataset/domain. Third, the V2 set is derived from a post-V1 source pool and has only 30 items. Fourth, historical model and processor revisions are not pinned. Fifth, related work and legal redistribution details still require verified sources. Diagnostic polarity and mechanism probes also contain format-specific parse failures and cannot support mechanistic claims.

The shortest credible path is to repair and hash-lock the execution notebook and transactional importer; complete blinded review before any new unblinding; use the 30-item runs only if their retrospective sensitivity value justifies the compute; build an independent control set; apply the frozen decision there; and only then decide whether a locked Main-500 or a small second-domain confirmation adds more value. The current local assets do not include unused control candidates or a second-domain dataset, so the independent set requires new data acquisition outside this local-only pass.

8 Conclusion

The real pilot shows large descriptive update gaps and model-dependent specificity under the frozen V1 rule. Stronger conclusions require blinded human review, pinned replication, and an independent powered control.